

Design a low-power **64-bit RISC-V compute subsystem** for **mobile XR** under a **3 W, 3 nm-class LP envelope**, returning a **DSE report with evidence and rejected alternatives**.

Intent

Improve mobile XR efficiency under power, reliability, memory, and deployment constraints.

Design space

Legal accelerator, memory, and vector choices; invalid power violations and deferred software work.

Environment

Simulator or cost model plus workload harness; actions change tiling, memory, or vector width.

Feedback budget

Many cheap proxies, fewer simulations, and rare high-fidelity checks before review.

Failed / rejected

Power violations, missed real-time bounds, bandwidth overruns, unsupported software, or proxy mismatches.

Commitment boundary

Authorize a higher-fidelity study only; no RTL, integration, or product commitment.

Task

Bounded DSE for a RISC-V compute subsystem, starting with an accelerator and memory slice.

Representation

XRBench traces, architecture parameters, compiler assumptions, power model, and uncertainties.

Method role

Generate candidates, predict cost, optimize, critique assumptions, and write evidence reports.

Evidence

Pareto comparison over latency, energy, area proxy, memory traffic, QoS, and sensitivity.

Rejection authority

Declared checks, constraint violations, QoS misses, cross-tool disagreement and independent review.

Accountable decision

The architect may refine, escalate fidelity, change scope, or reject the candidate.